

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03 COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: *Apply Dynamic Programming methods to solve reinforcement learning problems and derive optimal policies for decision-making tasks. (BL2, BL3, BL6).*

Assessment Tool Weightage: 25%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Data Interpretation

Duration: 1hr

1. A Reinforcement Learning agent is trained for multiple episodes and the reward values increase gradually from 10 to 45. Interpret the trend in rewards and defend what indicates about the learning performance and policy improvement of the agent. **(5 Marks)**
2. An epsilon-greedy agent is tested with different epsilon values such as 0.1, 0.3 and 0.5. The cumulative reward obtained is highest when epsilon is 0.1 and lowest when epsilon is 0.5. Justify and apply the concept of exploration vs exploitation to interpret why the reward changes with epsilon and identify which epsilon value provides better performance. **(5 Marks)**
3. Two Reinforcement Learning algorithms, Q-Learning and SARSA, are compared based on average rewards obtained over 100, 200 and 300 episodes. Q-Learning consistently shows slightly higher reward values than SARSA. Identify the

performance difference between these algorithms and determine which algorithm converges faster based on the observed results.

(5 Marks)

4. An RL agent is trained using different discount factor values such as 0.2, 0.5 and 0.9. The average reward increases as the discount factor increases. Discuss how the discount factor affects long-term reward optimization and justify which discount factor is most suitable for achieving better learning performance. **(5 Marks)**

5. The success rate of an RL agent increases from 45 percent at 50 episodes to 92 percent at 200 episodes. Discuss the improvement in success rate and interpret whether the agent has learned an optimal policy. Provide reasoning based on the trend of the data. **(5 Marks)**

6. An RL agent is trained in the CartPole environment with different learning rates (0.01, 0.05, 0.1). The agent achieves higher stability at 0.05 compared to the other values. Analyze how learning rate influences convergence and justify which value provides the best trade-off between speed and stability. **(5 Marks)**

7. A Q-learning agent is tested with different exploration strategies: ϵ -greedy, softmax, and random exploration. The cumulative reward is highest with ϵ -greedy. Compare the strategies and defend why ϵ -greedy achieves better performance in this scenario. **(5 Marks)**

8. An RL agent trained in a maze environment initially takes 50 steps to reach the goal, but after 300 episodes it reduces to 15 steps. Interpret the reduction in steps and evaluate whether this indicates convergence to an optimal policy. **(5 Marks)**

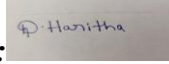
9. A SARSA agent is trained with discount factors of 0.4, 0.7, and 0.95. The agent achieves the highest long-term reward with 0.95. Discuss the role of discount factor in balancing immediate vs. future rewards and justify why 0.95 is most effective. **(5 Marks)**

10. An RL agent trained in a stochastic environment shows fluctuating rewards in early episodes but stabilizes after 500 episodes. Examine the reward trend and defend how stability indicates policy improvement and robustness of learning. **(5 Marks)**

Code of Conduct:

*I **Dasari Haritha(192425254)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



DATA INTERPRETATION

QUESTION 1 – Reward Trend and Policy Improvement

PROBLEM

A Reinforcement Learning agent is trained for multiple episodes, and the reward values gradually increase from **10 to 45**. Interpret the trend and explain what it indicates about the learning performance and policy improvement.

Given Data

Training Stage	Reward
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Initial Episode	10
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Episode 2	20
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Episode 3	30
-----------	----

Episode 4	40
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Final Episode	45
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1. RL CONCEPT

The important concept is **reward-based learning and policy improvement**.

In Reinforcement Learning, an agent interacts with an environment:

The agent learns to choose actions that maximize the **cumulative reward**.

2. OBSERVATION

The reward increases from:

Therefore, the overall reward trend is **positive and increasing**.

The total improvement is:

Percentage improvement:

Thus, the final reward is **350% higher than the initial reward**.

3. INTERPRETATION

Initially, the agent has little knowledge about the environment. Therefore, its actions are not optimal and it receives a low reward of **10**.

As training episodes increase, the agent learns from previous experiences. It gradually identifies actions that produce better outcomes.

The increasing reward indicates:

- Improved action selection
- Better policy
- Learning from experience
- Increased cumulative reward
- Improved interaction with the environment
- Reduction in poor decisions

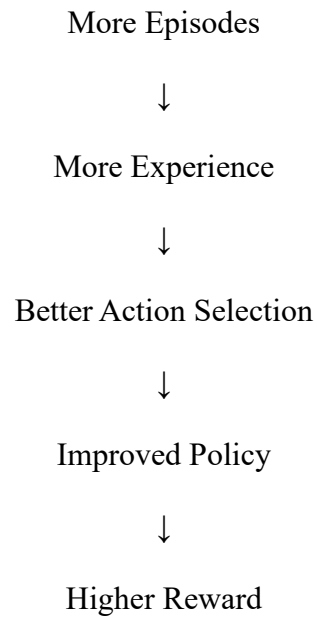
4. POLICY IMPROVEMENT

The policy is represented as:

It determines the probability of selecting action in state .

During training, the agent changes its policy so that actions producing higher rewards become more likely.

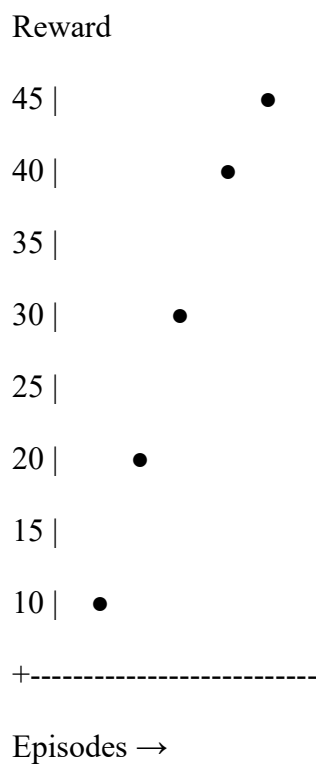
Therefore:



5. CONVERGENCE ANALYSIS

If the reward continues increasing and eventually becomes stable around a high value, it indicates that the agent may be approaching **policy convergence**.

For example:



The curve shows a positive learning trend.

6. IMPORTANT POINT

An increasing reward **does not automatically prove that the policy is globally optimal**.

To establish optimality, we should also check:

- Whether reward becomes stable
- Performance over additional episodes
- Average reward
- Environment-specific optimal reward
- Variance of rewards

CONCLUSION

The increase in reward from **10 to 45** demonstrates significant learning progress. The agent is improving its policy and selecting better actions as it gains experience. If the reward eventually stabilizes at a high level, it provides evidence that the policy is approaching convergence.

QUESTION 2 – Epsilon-Greedy Exploration vs Exploitation

PROBLEM

An epsilon-greedy agent is tested with ϵ values **0.1, 0.3 and 0.5**. The cumulative reward is highest at $\epsilon = 0.1$ and lowest at $\epsilon = 0.5$. Justify the result using exploration and exploitation.

Given Data

ϵ	Exploration Probability	Exploitation Probability	Cumulative Reward
0.1	10%	90%	Highest
0.3	30%	70%	Medium

ϵ	Exploration Probability	Exploitation Probability	Cumulative Reward
0.5	50%	50%	Lowest

1. RL CONCEPT

The relevant concept is the **exploration-exploitation trade-off**.

Exploration

The agent tries actions that may not currently be considered the best.

Exploitation

The agent selects the action currently believed to provide the highest reward.

2. EPSILON-GREEDY STRATEGY

The ϵ -greedy method works as follows:

Therefore:

For $\epsilon = 0.1$

The agent mainly exploits the best-known actions.

For $\epsilon = 0.5$

The agent spends much more time trying alternative actions.

3. INTERPRETATION

The highest cumulative reward occurs at $\epsilon = 0.1$.

This indicates that, in the given environment, the agent has already learned useful actions, so exploiting the learned policy produces better performance.

At $\epsilon = 0.5$, the agent explores too frequently. As a result, it selects less-effective actions more often, reducing cumulative reward.

4. COMPARISON

$\epsilon = 0.1$

Exploration ↓

Exploitation ↑

Reward ↑↑

↓

Best performance

$\epsilon = 0.3$

Exploration = Moderate

Exploitation = Moderate

Reward = Medium

$\epsilon = 0.5$

Exploration ↑

Exploitation ↓

Reward ↓

↓

Lowest performance

5. IMPORTANT ANALYSIS

A high exploration rate is not necessarily bad.

During **early training**, exploration is useful because the agent needs to discover good actions.

During **later training**, exploitation becomes more important because the agent can use its learned knowledge.

Therefore, many RL algorithms use **epsilon decay**:

Example:

Early training:

$\epsilon = 1.0 \rightarrow$ High exploration

Middle training:

$\epsilon = 0.5 \rightarrow$ Balanced

Late training:

$\epsilon = 0.1 \rightarrow$ High exploitation

CONCLUSION

Among the given values, $\epsilon = 0.1$ provides the best performance because it allows the agent to exploit its learned policy most of the time while still maintaining some exploration.

QUESTION 4 – Effect of Discount Factor

PROBLEM

An RL agent is trained with discount factors **0.2, 0.5 and 0.9**. The average reward increases as the discount factor increases. Discuss the effect of γ and identify the most suitable value.

Given Data

Discount Factor γ	Importance of Future Rewards	Average Reward
0.2	Very Low	Low
0.5	Moderate	Medium
0.9	High	Highest

1. RL CONCEPT

The discount factor controls the importance of **future rewards**.

The return is:

where:

- = return
- = reward
- = discount factor

The value normally satisfies the Bellman relationship:

2. $\gamma = 0.2$

A value close to zero means that the agent strongly prefers immediate rewards.

Immediate reward → Very important

Future reward → Less important

This is suitable for problems where immediate decisions are more important.

3. $\gamma = 0.5$

This gives a balance between immediate and future rewards.

Immediate reward → Important

Future reward → Moderately important

4. $\gamma = 0.9$

A high discount factor gives substantial importance to future rewards.

Immediate reward → Important

Future reward → Very important

The agent is encouraged to make decisions that may produce lower immediate rewards but higher long-term returns.

5. INTERPRETATION OF DATA

The average reward increases as:

This indicates that the environment benefits from **long-term planning**.

The agent performs better when it considers future consequences.

6. EXAMPLE

Suppose two actions are available:

Action A:

Immediate reward = 10

Future reward = 0

Action B:

Immediate reward = 5

Future reward = 20

With a high γ , the agent can recognize the long-term benefit of Action B.

Therefore, high γ is useful for sequential tasks.

7. INTERPRETATION

The data clearly indicates:

Since $\gamma = 0.9$ produces the highest average reward, it is the best among the tested values.

CONCLUSION

The discount factor strongly influences the agent's decision-making strategy. In this experiment, $\gamma = 0.9$ is most suitable because the environment requires long-term planning and future rewards contribute significantly to successful behavior.

QUESTION 6 – Learning Rate in CartPole

PROBLEM

An RL agent is trained in CartPole using learning rates **0.01**, **0.05** and **0.1**. The agent achieves the highest stability at **0.05**. Analyze the effect of learning rate and justify the best value.

Given Data

Learning Rate	Learning Behavior	Stability
0.01	Very slow updates	Moderate
0.05	Balanced updates	Highest
0.10	Large updates	Lower

1. RL CONCEPT

The **learning rate**, usually represented by α , controls how much the agent changes its parameters during each update.

A simplified update is:

2. LEARNING RATE = 0.01

This is a small learning rate.

Advantages:

- Small parameter updates
- More gradual learning

Disadvantage:

- Slow convergence
- Requires more training episodes

$$\alpha = 0.01$$



Small updates



Slow learning

3. LEARNING RATE = 0.10

This is a relatively large learning rate.

Advantages:

- Faster initial learning

Disadvantages:

- Large updates
- Possible overshooting
- Oscillation
- Instability

$$\alpha = 0.10$$



Large updates



Possible oscillation



Lower stability

4. LEARNING RATE = 0.05

This provides a balance.

$$\alpha = 0.05$$



Moderate updates



Good learning speed



Stable convergence

Therefore, the observed highest stability at **0.05** indicates that it is the best trade-off for this experiment.

5. COMPARISON

0.01 —————▶ Stable but slow

0.05 —————▶ Fast + Stable ★ BEST

0.10 —————▶ Fast but unstable

6. WHY STABILITY MATTERS

In CartPole, the agent must continuously balance the pole.

If the learning rate is too large, the policy can change drastically from one update to another, causing inconsistent behavior.

A moderate learning rate allows the agent to gradually improve the control policy.

CONCLUSION

The learning rate **0.05** provides the best trade-off between learning speed and stability. The value is neither too small nor too large, allowing the agent to learn efficiently while maintaining stable convergence.

QUESTION 8 – Maze Learning and Convergence

PROBLEM

An RL agent initially takes **50 steps** to reach the goal in a maze. After 300 episodes, it takes only **15 steps**. Interpret the reduction and determine whether it indicates convergence to an optimal policy.

Given Data

Training Stage	Steps to Goal
Initial	50
After 300 Episodes	15

1. OBSERVATION

The number of steps decreases from:

Reduction:

Therefore, the agent saves **35 steps**.

2. PERCENTAGE IMPROVEMENT

Thus, the number of steps has been reduced by **70%**.

3. INTERPRETATION

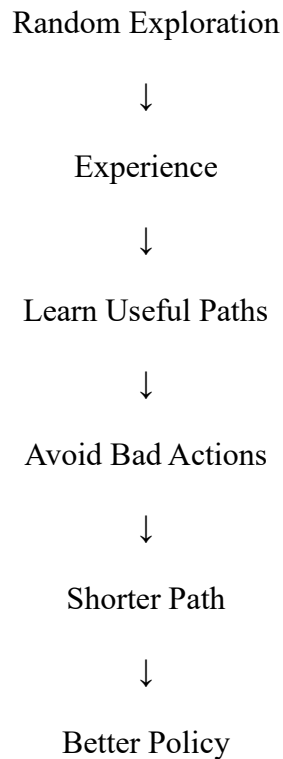
Initially, the agent does not know the optimal route.

It may:

- Take unnecessary paths
- Explore different directions
- Hit dead ends
- Return to previously visited locations

After many episodes, it learns which actions lead toward the goal.

The learning process becomes:



4. POLICY IMPROVEMENT

The reduction in steps indicates that the policy has improved.

The agent is now selecting actions that lead to the goal more efficiently.

Therefore:

5. DOES IT PROVE OPTIMALITY?

Not necessarily.

The result strongly indicates **learning and policy improvement**, but 15 steps can only be called optimal if the shortest possible path in the maze is also 15 steps.

For example:

Shortest possible path = 15

Agent's path = 15



Optimal policy likely

But:

Shortest possible path = 10

Agent's path = 15



Policy is improved but NOT optimal

6. CONVERGENCE

To confirm convergence, we should observe the agent over additional episodes.

For example:

Episode	Steps
---------	-------

300	15
-----	----

350	15
-----	----

400	15
-----	----

450	15
-----	----

500	15
-----	----

If the steps remain around 15, the policy has likely converged.

CONCLUSION

The reduction from **50 to 15 steps** represents a **70% improvement** and strongly indicates that the agent has learned an efficient policy. However, the data alone does not prove that the policy is optimal. Optimality requires comparison with the shortest possible path and observation of stable performance over additional episodes.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand